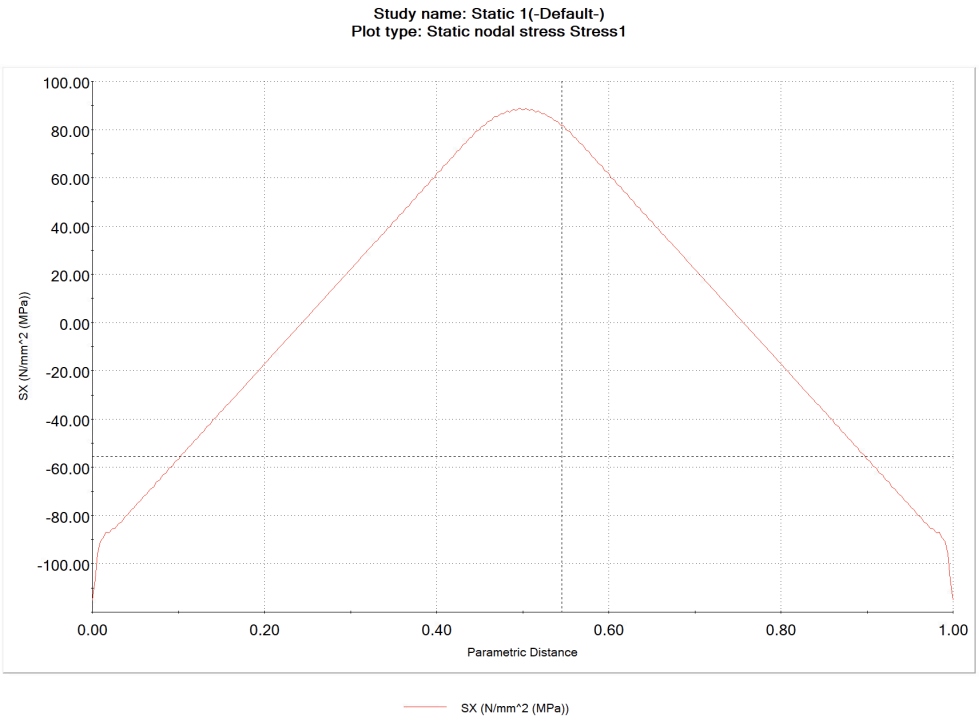
Modeling a Pull-Up Bar in FEA

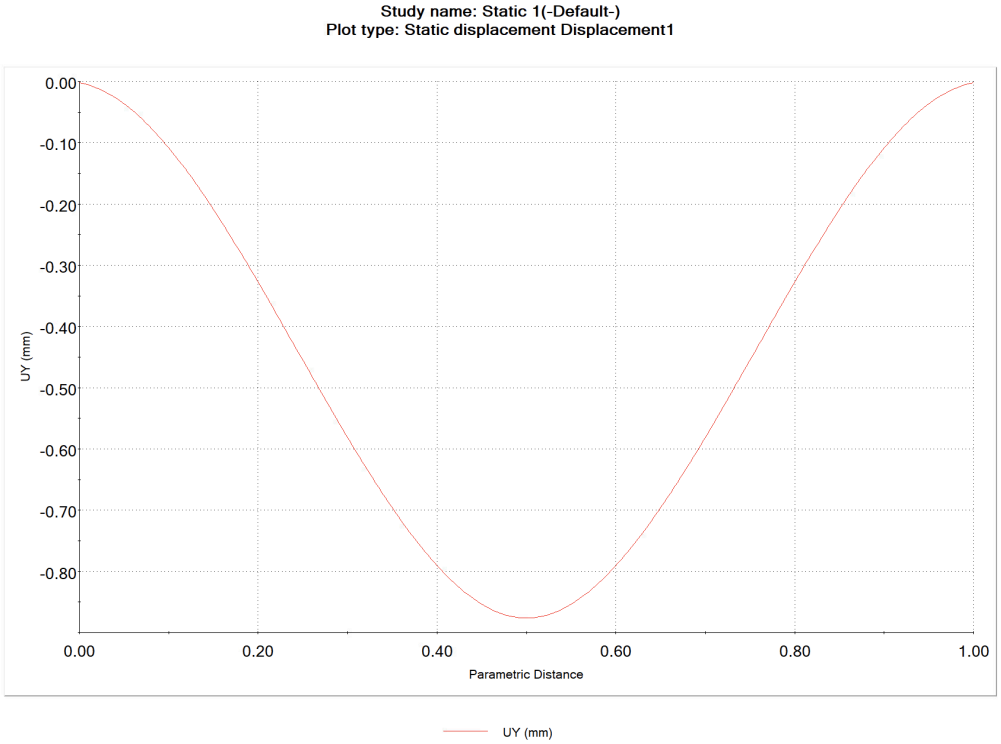
Anne Jumper

Deliverable 1: Plot of the normal stress in the X direction (SX) along the bottom of the bar.



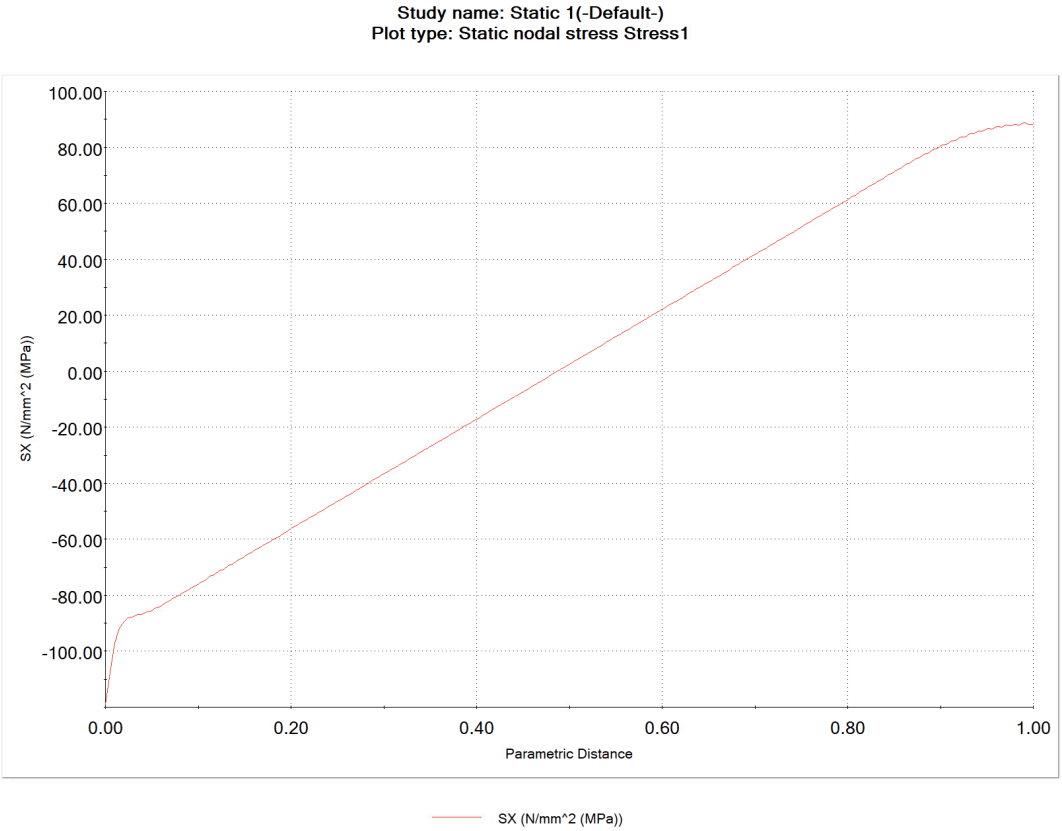
Peak Normal Stress = 89 MPa

Deliverable 2: Provide a plot of the displacement in the Y direction (UY) along the bottom of the bar.



Peak Displacement = -0.88mm

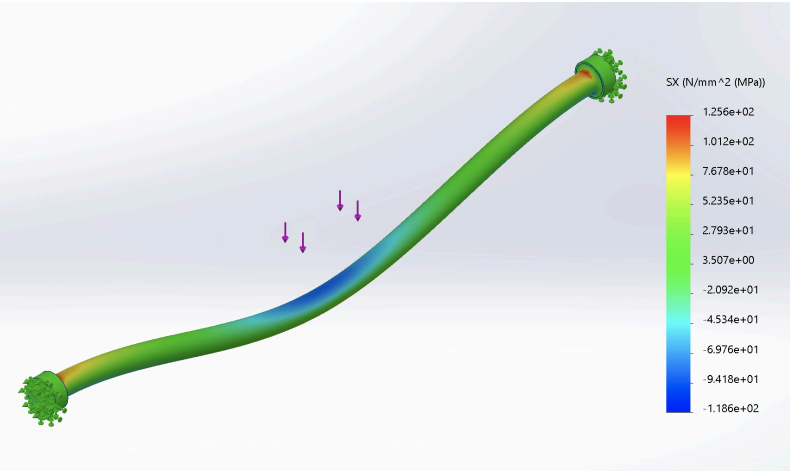
Deliverable 3: Provide a plot of the normal stress in the X direction (SX) along the bottom of the half bar.



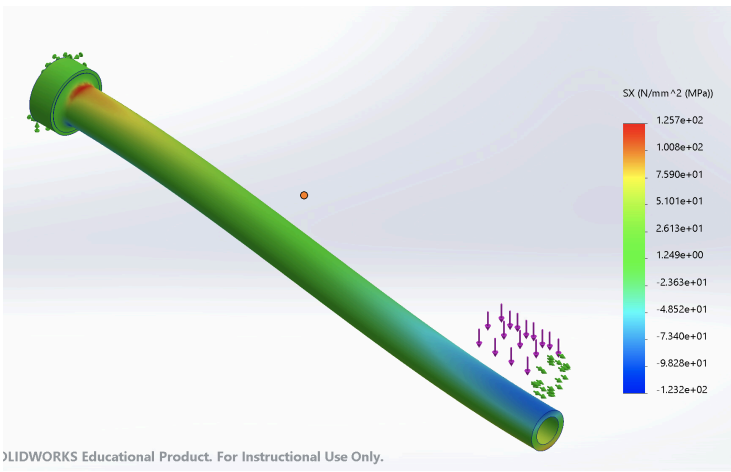
Peak Normal Stress = 89MPa
The peak normal stress is the same in both the half and full bar as seen in the probe results.

However, we can see that the stress (negative, blue) at the center of the half bar is higher than that in the center of the full bar and that the stress at the end of the bar (positive, red) is slightly higher for the half bar than for the full bar.

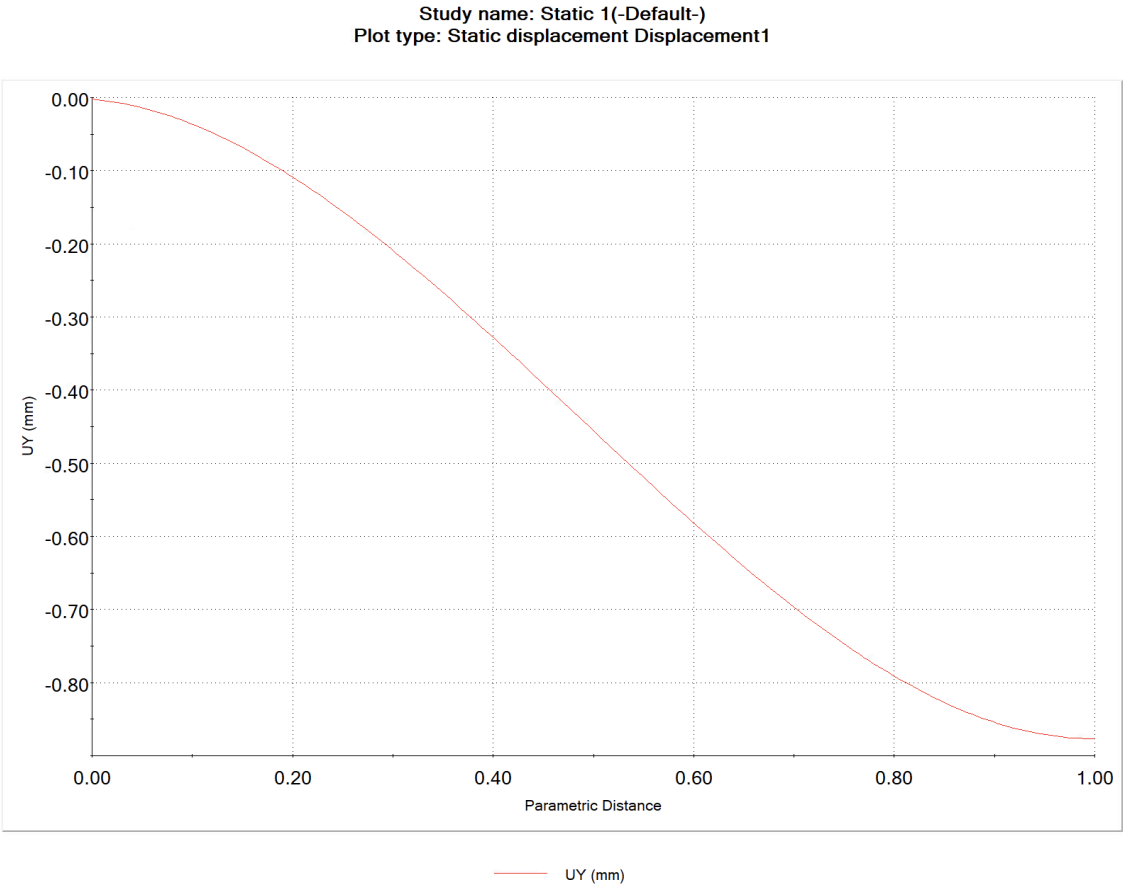
Full Bar



Half Bar



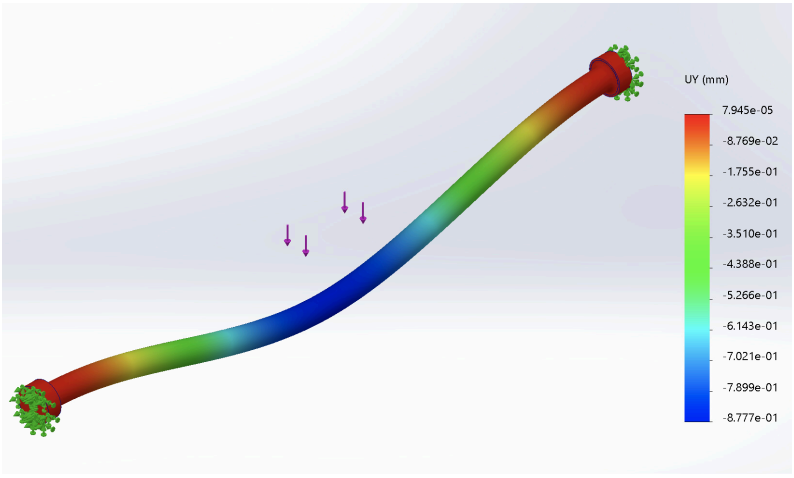
Deliverable 4: Provide a plot of the displacement in the Y direction (UY) along the bottom of the half bar.



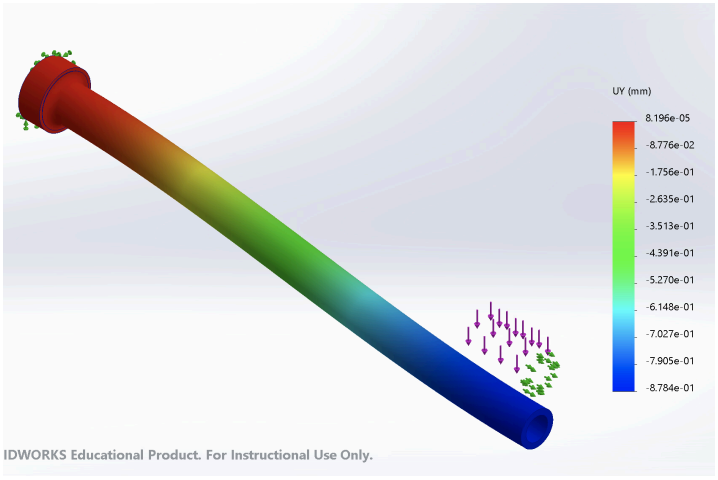
Peak Displacement = -0.88 mm
The peak displacement is the same for both the half and full bars as seen in the probe results

However, looking at the displacement plots, we can see there is slightly more displacement in the half bar at the center (blue, negative displacement) and slightly more displacement at the end of the half bar (red, positive displacement).

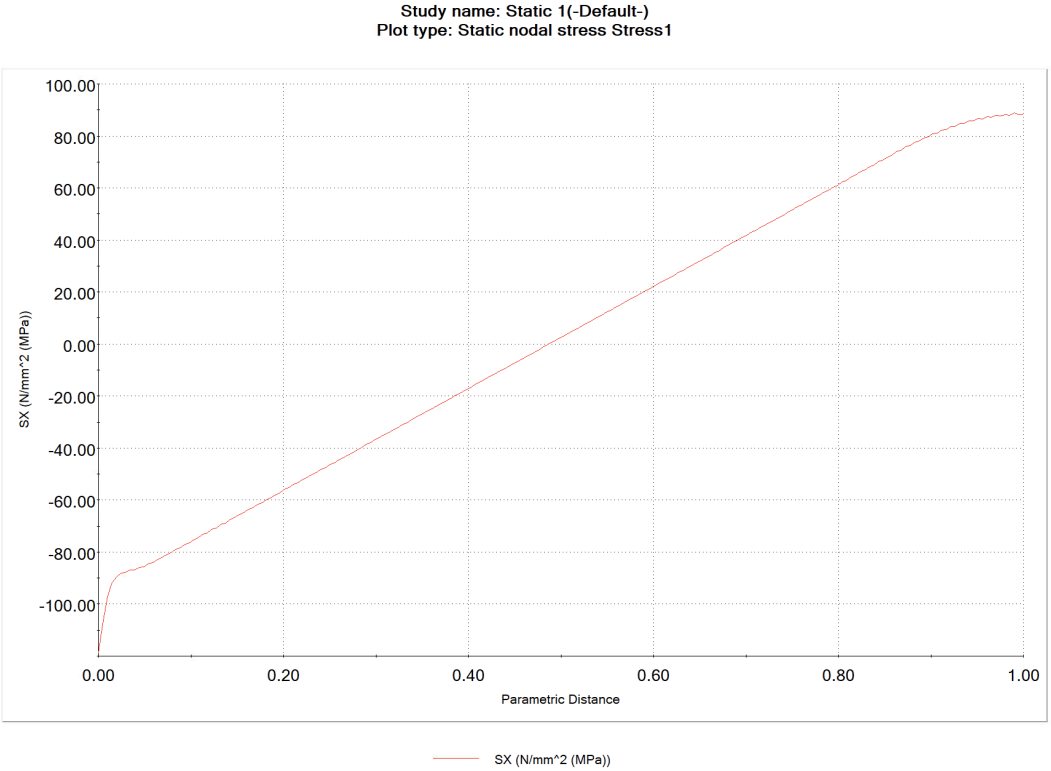
Full Bar



Half Bar

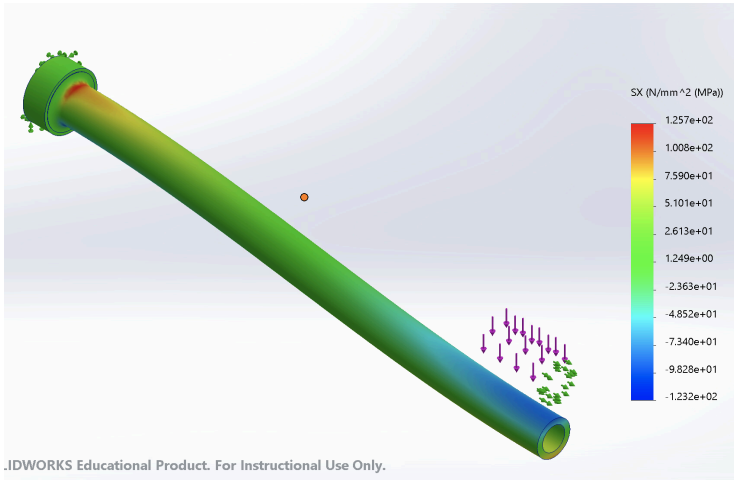


Deliverable 5: Provide a plot of the normal stress in the X direction (SX) along the bottom of the half bar (with foundation bolts).

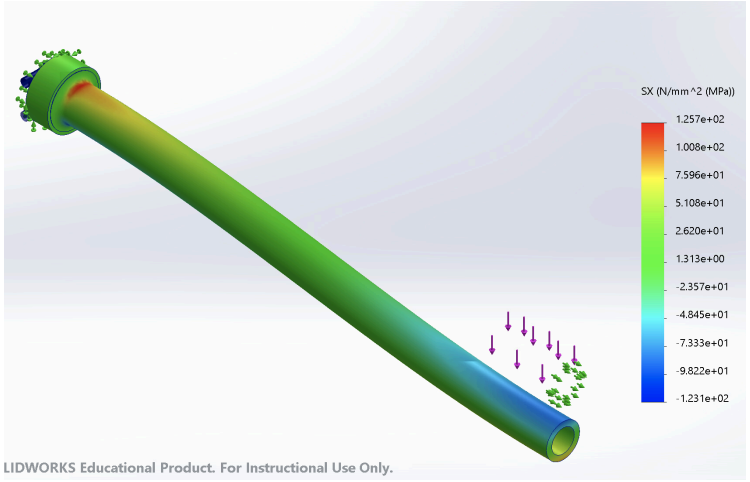


No significant differences in the results of the probing, however, the peak normal stress of the half-bar with the fixed condition reaches a higher magnitude at the center of the bar (blue, negative stress) than for the half-bar with the foundation bolts.

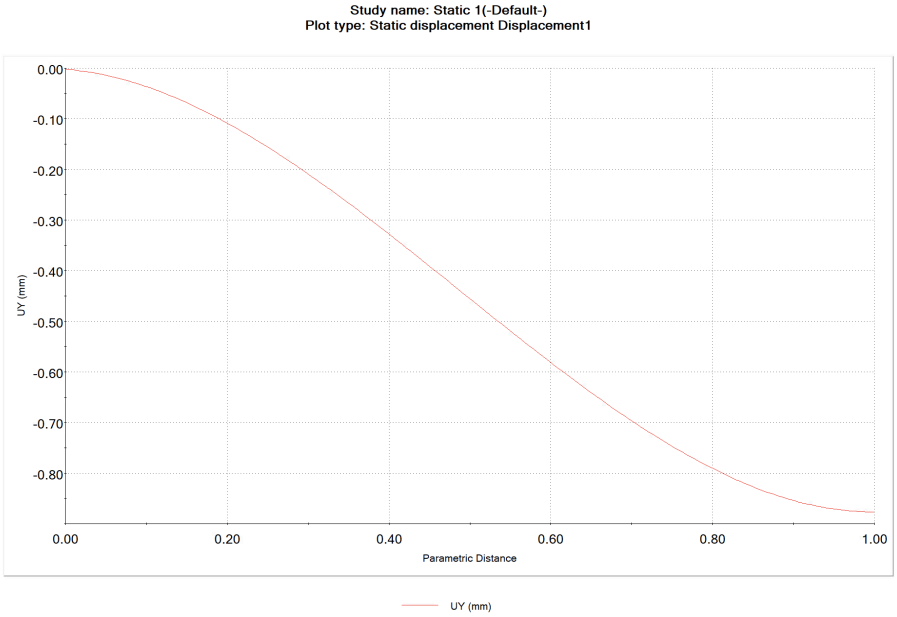
Fixed Condition



Foundation Bolts



Deliverable 6: Provide a plot of the displacement in the Y direction (UY) along the bottom of the half bar (with foundation bolts).



No significant differences in the results of the probing, however, the peak displacement of the half-bar with the fixed condition reaches a higher magnitude at the center of the bar (blue, downward/negative displacement) than for the bar with the bolts.

Fixed Condition

